Session 4 | Linear Algebra II

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Keep tying eigenvalues back to their two big uses: PCA and the stability of dynamical systems.

Exercise 1 $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

Solution.

(a) $\boldsymbol{A}(1,1) = (4+1,\ 2+3) = (5,5) = 5\,(1,1)$, so $(1,1)$ is an eigenvector with eigenvalue $\boxed{5}$.

(b) An eigenvector is a direction the matrix only *stretches* (by the eigenvalue), without rotating it off its own line.

Exercise 2 *Eigenvalues of* $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

Solution. $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (4-\lambda)(3-\lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$. Factoring, $(\lambda-5)(\lambda-2) = 0$, so $\lambda = \boxed{5 \text{ and } 2}$.

Reminder

For a 2×2 matrix the characteristic equation is always $\lambda^2 - (\text{trace})\lambda + \det = 0$.

Exercise 3 *Eigenvector for* $\lambda = 2$.

Solution. $\boldsymbol{A} - 2\boldsymbol{I} = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$, giving $2x + y = 0$, i.e. $y = -2x$. An eigenvector is $\boxed{(1,-2)}$.

Exercise 4 *Trace/determinant checks.*

Solution. $\text{trace} = 4 + 3 = 7 = 5 + 2 = \lambda_1 + \lambda_2$; $\quad \det = 4(3) - 1(2) = 10 = 5 \cdot 2 = \lambda_1 \lambda_2$. Both check out.

Reminder

Always verify eigenvalues with $\sum \lambda_i = \text{trace}$ and $\prod \lambda_i = \det$, a free way to catch algebra slips.

Exercise 5 $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ *(symmetric).*

Solution.

(a) $(3-\lambda)^2 - 1 = 0 \Rightarrow 3 - \lambda = \pm 1 \Rightarrow \lambda = \boxed{4 \text{ and } 2}$.

(b) $\lambda = 4$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \Rightarrow (1,1)$; $\lambda = 2$: $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) \Rightarrow (1,-1)$. Check $(1,1)\cdot(1,-1) = 0$: orthogonal.

(c) Symmetric matrices are guaranteed real eigenvalues and orthogonal eigenvectors (the spectral theorem).

Reminder

Every covariance matrix is symmetric, so its principal axes are always orthogonal, exactly what makes PCA well-behaved.

Exercise 6 *Discrete system; eigenvalues* 0.9 *(dir* $\boldsymbol{u}$*) and* 0.5 *(dir* $\boldsymbol{v}$*), each coefficient* 1.

Solution. Along an eigenvector, one step multiplies the coefficient by λ.

(a) After 1 step: $\boxed{0.9 \text{ and } 0.5}$.

(b) After 10 steps: $0.9^{10} \approx \boxed{0.349}$ and $0.5^{10} \approx \boxed{0.001}$.

(c) The $\lambda = 0.9$ mode dominates: it decays far more slowly, so after many steps the state points essentially along $\boldsymbol{u}$.

Reminder

$\boldsymbol{A}^k$ multiplies each eigen-coefficient by λ^k. The eigenvalue of largest magnitude sets the long-run behaviour.

Exercise 7 $\boldsymbol{C} = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 5 \end{smallmatrix}\right)$.

Solution.

(a) $(5-\lambda)^2 - 4 = 0 \Rightarrow 5 - \lambda = \pm 2 \Rightarrow \lambda = \boxed{7 \text{ and } 3}$.

(b) PC1 (for $\lambda = 7$): $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \Rightarrow (1,1)$.

(c) Fraction $= \dfrac{7}{7+3} = \boxed{70\%}$.

Reminder

PCA *is* the eigen-decomposition of the covariance matrix: eigenvectors are the principal components, eigenvalues are the variances along them.

Exercise 8 *Eigenvalues* $12, 6, 1.5, 0.5$.

Solution.

(a) Total $= 12 + 6 + 1.5 + 0.5 = 20$; PC1 explains $\dfrac{12}{20} = \boxed{60\%}$.

(b) Cumulative: $\frac{12}{20} = 60\%$, then $\frac{12+6}{20} = 90\%$. So $\boxed{\text{2 components}}$ reach $\geq 90\%$.

Exercise 9 *Eigenvalues* $8, 5, 2$.

Solution.

(a) Total variance $= 8 + 5 + 2 = \boxed{15}$; it equals the *trace* of the covariance matrix.

(b) No: a covariance matrix is positive-semidefinite, so all eigenvalues are ≥ 0 (a variance cannot be negative).

Reminder

Total variance $= \sum \lambda_i = \mathrm{trace}(\boldsymbol{C})$, and every eigenvalue of a covariance matrix is non-negative.

Exercise 10 *Could these be covariance matrices?*

Solution.

(a) Eigenvalues $3, 1$: $\boxed{\text{Yes}}$, both positive (positive-definite).

(b) Eigenvalues $4, -2$: $\boxed{\text{No}}$, a negative eigenvalue is impossible.

(c) Eigenvalues $2, 0$: $\boxed{\text{Yes, but degenerate}}$, positive-semidefinite, with zero variance along one direction (the data lie on a line).

Reminder

"Valid covariance matrix" $\Leftrightarrow$ symmetric and positive-semidefinite (all $\lambda \geq 0$). A zero eigenvalue signals a redundant, rank-deficient dataset.

Exercise 11 *Recurrent network, eigenvalues* 0.95 *and* 1.05.

Solution.

(a) $\boxed{\text{Unstable}}$: $|1.05| > 1$.

(b) Activity grows without bound along the 1.05 mode (the 0.95 mode decays).

(c) If both were 0.95: $\boxed{\text{stable}}$, every $|\lambda| < 1$, so activity decays to zero.

Reminder

Discrete-time stability requires *every* eigenvalue to satisfy $|\lambda| < 1$. A single mode with $|\lambda| > 1$ makes the whole system blow up.

Exercise 12 *Continuous system, eigenvalues* $-1 \pm 2i$.

Solution.

(a) $\boxed{\text{Stable}}$: the solution behaves like $e^{(-1\pm 2i)t}$, and the *real part* $-1 < 0$ makes the amplitude decay.

(b) The imaginary part $\pm 2i$ produces *oscillations*; here, damped oscillations that spiral in toward the equilibrium.

> **Reminder**
>
> Continuous-time stability requires every eigenvalue to have negative real part. Imaginary parts set the oscillation frequency; real parts set growth or decay.

Exercise 13 *Continuous system, eigenvalues* 2 *(dir* $\boldsymbol{u}$*) and* -3 *(dir* $\boldsymbol{v}$*).*

Solution. Along $\boldsymbol{u}$ the state grows like e^{2t}; along $\boldsymbol{v}$ it decays like e^{-3t}. After a long time the state aligns with $\boxed{\boldsymbol{u} \text{ and grows without bound}}$.

Exercise 14 100 *neurons; first* 3 *PCs explain* 85%.

Solution.

(a) The population activity is effectively *low-dimensional*: roughly 3 directions capture most of it, even though 100 neurons were recorded.

(b) Those few components can be plotted and analysed, and dropping the rest removes noise (denoising / visualisation).

(c) Variance is not the same as *relevance*: a direction carrying little variance may still be the one that matters for behaviour.

> **Reminder**
>
> Low-dimensional structure is ubiquitous in neural data, but PCA ranks directions by variance, not by how informative they are about the task.

Exercise 15 *Symmetric* $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{\top}$.

Solution.

(a) The columns of $\boldsymbol{P}$ are the (orthonormal) *eigenvectors* of $\boldsymbol{A}$.

(b) The diagonal entries of $\boldsymbol{D}$ are the *eigenvalues*.

(c) Applying $\boldsymbol{A}$ rotates space onto the eigenvector axes, stretches each axis by its eigenvalue, then rotates back, independent stretches along perpendicular directions.

> **Reminder**
>
> This spectral decomposition is why a symmetric matrix is "just" a set of scalings along orthogonal axes, the geometric picture behind PCA and the SVD.